RESOLVING A CATEGORY ERROR: DHL$[3,2]$ YIELDS $H_1 \le 6$, NOT $H_1 = 2$

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ABSTRACT. Let $H_1 = \liminf_{n\to\infty}(p_{n+1}-p_n)$. We record the precise logical relationship between the hypothesis DHL$[3,2]$ and bounds on H_1. We prove that DHL$[3,2]$ implies $H_1 \le 6$; that $H_1 \le 3$ is equivalent to $H_1 = 2$ (the twin prime conjecture); that the minimal diameter of an admissible 3-tuple is exactly 6; and, via an explicit model, that DHL$[2,2]$ cannot be derived from the combinatorial form of DHL$[3,2]$ alone. The parity barrier for square-of-divisor-sum sieves enters only as an explicitly stated hypothesis. We do not prove DHL$[3,2]$ or the twin prime conjecture. All results are formalized in Lean 4 with Mathlib, without `sorry` and using only Lean's standard axioms. This is the final version of the note: every result is stated to match a named Lean declaration, whose file and line are given in Section 5. The main result is conditional on DHL$[3,2]$, an open conjecture, and the paper does not remove that hypothesis.

1. INTRODUCTION

Let $p_1 = 2 < p_2 = 3 < \dots$ denote the primes and $H_1 = \liminf_{n\to\infty}(p_{n+1} - p_n)$. For an admissible k-tuple $\mathcal{H}$ (Definition 2.1), the statement DHL$[k, j]$ asserts that infinitely many translates $n + \mathcal{H}$ contain at least j primes (Definition 2.2).

Following Goldston, Pintz and Yıldırım [2], Maynard [3] and the Polymath8b project [5], the unconditional bound $H_1 \le 246$ is known, and DHL$[3, 2]$ is known conditionally on the generalized Elliott–Halberstam conjecture (GEH), yielding $H_1 \le 6$ under GEH [5]. A claimed improvement to $H_1 \le 240$, attributed to a 2026 preprint of Stadlmann, is **[citation needed: unverified]** and is not used here.

The purpose of this note is to separate two statements that are sometimes conflated: DHL$[3, 2] \Rightarrow H_1 \le 6$ (Theorem 3.2), which is elementary, and $H_1 = 2$, the twin prime conjecture. Since every prime gap after $3 - 2$ is even (Theorem 3.1), $H_1 \le 3$ is equivalent to $H_1 = 2$, so reading DHL$[3, 2]$ as giving $H_1 \le 3$ is a category error.

By the widely accepted parity barrier (going back to Selberg), standard square-of-divisor-sum sieves are not expected to go below 6, even under GEH; in this note that barrier is used only as an explicit hypothesis (Hypothesis 3.5), not proved.

2. PRELIMINARIES

Definition 2.1. A finite set $\mathcal{H} \subset \mathbb{N}$ is *admissible* if for every prime q there is a residue class modulo q containing no element of $\mathcal{H}$.

Definition 2.2. For a predicate P on $\mathbb{N}$ and $k \ge j \ge 1$, let DHL$_P[k, j]$ be the statement: for every admissible $\mathcal{H}$ with $|\mathcal{H}| = k$ there are infinitely many n such that $\#\{h \in \mathcal{H} : P(n+h)\} \ge j$. Write DHL$[k, j]$ for DHL$_P[k, j]$ with $P =$ primality.

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Definition 2.3. For $c \in \mathbb{N}$ write $H_1 \leq c$ for the statement that $p_{n+1} - p_n \leq c$ for infinitely many n. For an integer-valued sequence this is equivalent to $\liminf_n (p_{n+1} - p_n) \leq c$.

Lemma 2.4. *The sets* $\{0, 2\}$, $\{0, 2, 6\}$ *and* $\{0, 4, 6\}$ *are admissible.*

Proof. For $q \geq 7$ the elements are distinct residues below q and the class 1 is missed. For $q = 2$ the class 1 is missed. For $q = 3$: $\{0, 4, 6\} \equiv \{0, 1, 0\}$ misses 2; $\{0, 2, 6\} \equiv \{0, 2, 0\}$ and $\{0, 2\}$ miss 1. For $q = 5$: $\{0, 4, 6\} \equiv \{0, 4, 1\}$ misses 2; $\{0, 2, 6\} \equiv \{0, 2, 1\}$ misses 3; $\{0, 2\}$ misses 1. □

3. Main results

Theorem 3.1. *For all* $n \geq 2$, $p_{n+1} - p_n \geq 2$ *and* $p_{n+1} - p_n$ *is even. Consequently* $H_1 \geq 2$, *and* $H_1 \leq 3$ *if and only if* $p_{n+1} - p_n = 2$ *for infinitely many* n.

Proof. For $n \geq 2$, p_n and p_{n+1} are odd and distinct, so their difference is a positive even integer. Hence a gap ≤ 3 with $n \geq 2$ equals 2; the only gap equal to 1 is $p_2 - p_1$. □

Theorem 3.2. $\mathrm{DHL}[3, 2]$ *implies* $H_1 \leq 6$.

Proof. By Lemma 2.4, $\{0, 4, 6\}$ is admissible, so $\mathrm{DHL}[3, 2]$ gives infinitely many n such that at least two of $n, n+4, n+6$ are prime. For each such n we obtain primes $p < q$ with $q - p \in \{2, 4, 6\}$ and $p \geq n$. Write $p = p_k$. Since q is a prime larger than p, the next prime satisfies $p_{k+1} \leq q$, so $p_{k+1} - p_k \leq q - p \leq 6$. As $p \geq n$ and n is unbounded, so is k. □

Theorem 3.3. *Let* $P(n)$ *hold if and only if* $n \bmod 4 \in \{0, 1\}$ *(in Lean* `Ptoy n := n % 4 < 2`*; the paper's wording is the kernel-checked equivalent* `Ptoy_iff_mod_four_mem`*). Then* $\mathrm{DHL}_P[3, 2]$ *holds (indeed for every* 3*-element set), but* $P(n) \wedge P(n+2)$ *holds for no* n*; in particular* $\mathrm{DHL}_P[2, 2]$ *fails.*

Proof. Integers differing by 2 have residues differing by 2 modulo 4, so they are never both in P. If $d \not\equiv 2 \pmod 4$, then $P(m) \wedge P(m + d)$ for all $m \equiv 0 \pmod 4$ when $d \equiv 0, 1$, and for all $m \equiv 1 \pmod 4$ when $d \equiv 3$. For $a < b < c$, the differences $b - a$ and $c - b$ cannot both be $\equiv 2 \pmod 4$, since then $c - a \equiv 0$. So some pair in the set has a difference $\not\equiv 2 \pmod 4$ and is hit infinitely often. Finally, $\{0, 2\}$ is admissible by Lemma 2.4. □

Remark 3.4. Theorem 3.3 shows only that no proof of $\mathrm{DHL}[3, 2] \Rightarrow \mathrm{DHL}[2, 2]$ can rely on the combinatorial form of $\mathrm{DHL}[3, 2]$ alone. It is *not* an independence result for the primes, and $\mathrm{DHL}[3, 2]$ is compatible with $H_1 = 2$.

Hypothesis 3.5 (Parity barrier)**.** A given sieve method establishes $\mathrm{DHL}[k, j]$ only if $j \leq \lceil k/2 \rceil$.

This hypothesis reflects the parity discussion in [5] for square-of-divisor-sum weights. It is assumed, not proved.

Theorem 3.6.

(1) *Under Hypothesis 3.5, a sieve method establishing* $\mathrm{DHL}[k, 2]$ *has* $k \geq 3$.
(2) *(Unconditional.) No admissible* 3*-element set is contained in an interval* $[a, a + 5]$. *Together with Lemma 2.4, the minimal diameter of an admissible* 3*-set is exactly* 6.

Proof. (1) $2 \leq \lceil k/2 \rceil$ forces $k \geq 3$. (2) Admissibility modulo 2 forces the three elements to share a parity. If they lie in an interval of length 5, any two differ by 2 or 4, so they are pairwise distinct modulo 3 and cover all residues modulo 3, contradicting admissibility modulo 3. □

4. Formal verification

All results above are formalized in Lean 4 with Mathlib [6] in the repository https://github.com/BenFrohman/DHL-3-2-implies-liminf-prime-gap-le-6. The development contains no `sorry`; `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. Primes are indexed from 0 in Lean ($p_0 = 2$). Key definitions and statements (rendered here in ASCII notation; the source uses Lean's Unicode symbols):

```
def H1Le (c : Nat) : Prop := forall N, exists n, n >= N /\ gap n <= c
def Admissible (H : Finset Nat) : Prop :=
  forall q, q.Prime -> exists r < q, forall h in H, h % q != r
def DHL (k j : Nat) : Prop :=
  forall H : Finset Nat, H.card = k -> Admissible H ->
    forall N, exists n, n >= N /\ j <= (H.filter (fun h => (n + h).Prime)).card

theorem dhl32_H1Le_six (h : DHL 3 2) : H1Le 6
theorem H1Le_three_iff_twin : H1Le 3 <-> forall N, exists n, n >= N /\ gap n = 2
theorem dhl32_not_implies_dhl22_combinatorially :
  exists (P : Nat -> Prop) (_ : DecidablePred P),
    DHLfor P 3 2 /\ (forall n, Not (P n /\ P (n + 2))) /\ Not DHLfor P 2 2
structure ParityBarrier (Proves : Nat -> Nat -> Prop) : Prop where
  bound : forall k j, Proves k j -> j <= (k + 1) / 2
```

Correspondence with the text:

- Lemma 2.4: `admissible_02`, `admissible_026`, `admissible_046`.
- Theorem 3.1: `two_le_gap`, `not_H1Le_one`, `H1Le_three_iff_two`, `H1Le_three_iff_twin`.
- Theorem 3.2: `dhl32_H1Le_six`.
- Theorem 3.3: `Ptoy_DHL32`, `dhl32_not_implies_dhl22_combinatorially`.
- Theorem 3.6: `ParityBarrier`, `parity_two_needs_three`, `admissible_three_diam_ge_six`, `min_diam_admissible_three`.

Statements from the literature (the bound 246, the conditional status of DHL[3, 2], and the unverified Stadlmann claim) are not formalized.

5. Formal statements, provenance and axioms

Exact correspondence. Each result above is the informal reading of the Lean declaration listed below. Line numbers refer to the Lean sources at commit `9a9f97d` (merge of `pipelinehalt/align-wording` into the default branch); the Lean sources are unchanged in tag `v1.4`.

Paper	Lean declaration	File:line
Definition 2.3	`H1Le`	`DHL32/Basic.lean:18`
Definition 2.1	`Admissible`	`DHL32/Basic.lean:22`
Definition 2.2	`DHL`; `DHLfor`	`Basic.lean:27`; `Theorems34.lean:17`
Lemma 2.4	`admissible_046`, `admissible_026`	`Basic.lean:97, 113`
	`admissible_02`	`Theorems34.lean:74`
Theorem 3.1	`two_le_gap`, `gap_even`	`Basic.lean:45, 53`
	`not_H1Le_one`, `H1Le_three_iff_two`	`Basic.lean:62, 68`
	`H1Le_three_iff_twin`	`Basic.lean:81`
Theorem 3.2	`dhl32_H1Le_six`	`Basic.lean:148`
Theorem 3.3	`Ptoy`, `Ptoy_iff_mod_four_mem`	`Theorems34.lean:25, 30`
	`Ptoy_DHL32`	`Theorems34.lean:52`
	`dhl32_not_implies_dhl22_combinatorially`	`Theorems34.lean:94`
Hypothesis 3.5	`ParityBarrier` (a `structure`, not an axiom)	`Theorems34.lean:106`
Theorem 3.6	`parity_two_needs_three`	`Theorems34.lean:110`
	`admissible_three_diam_ge_six`, `min_diam_admissible_three`	`Theorems34.lean:116, 136`

The main theorem reads, verbatim in ASCII, `theorem dhl32_H1Le_six (h : DHL 3 2) : H1Le 6`. Its only hypothesis is DHL$[3, 2]$.

Indexing. Lean indexes primes from 0 (`p 0 = 2`); the paper indexes from 1 ($p_1 = 2$). The paper's "$n \geq 2$" in Theorem 3.1 is therefore Lean's `1 <= n` in `two_le_gap` and `gap_even`. This is the same statement, not a weakening.

Build and axioms. Toolchain `leanprover/lean4:v4.30.0-rc1`; Mathlib tag `v4.30.0-rc1` (commit `0c154d6`). On 3 November 2026 a full `lake build` of commit `9a9f97d` completed successfully. The Lean sources contain no `sorry`, no `native_decide` and no `axiom` declarations. `lake env lean AxiomAudit.lean` (`#print axioms`) reports, for every declaration, a subset of `[propext, Classical.choice, Quot.sound]`; the full output is in Appendix A.

Scope: what is and is not proved. Proved: the implication DHL$[3, 2] \Rightarrow H_1 \leq 6$, with DHL$[3, 2]$ as the explicit hypothesis `h : DHL 3 2`; and the unconditional statements Lemma 2.4, Theorem 3.1, Theorem 3.3 and Theorem 3.6(2). **Not proved:** DHL$[3, 2]$ itself, which is an open conjecture (known only conditionally on GEH [5]); any unconditional bound on H_1, bounded gaps, or the twin prime conjecture; the parity barrier, which enters only as the hypothesis `ParityBarrier`. Theorem 3.3 is not an independence result for the primes. Literature statements cited in Section 1 are not formalized.

6. Paper-to-Lean alignment check

The statements of this paper were compared with the Lean declarations using PipelineHalt, a paper-to-Lean alignment harness written by the author. PipelineHalt compares wording and structure: it flags places where a Lean statement appears narrower, conditional or silent relative to the paper, and halts with a question for the author rather than guessing. It does not verify mathematics; correctness is established only by the Lean kernel. On this repository it raised three flags: docstring numbering ("Theorem 1") versus the paper's numbering, closed by a hand-written mapping file; the index shift $n \geq 1$ versus $n \geq 2$, closed because the definitions settle it (`p 0 = 2`); and the residue wording $\{0, 1\}$ versus `n % 4 < 2`, closed by the kernel-checked lemma `Ptoy_iff_mod_four_mem`. No flags remain open. A clean alignment report means only that no unexplained wording-level difference was found.

7. Discussion

A proof of $H_1 = 2$ must go beyond the combinatorial content of DHL[3, 2] (Remark 3.4) and, if it is sieve-theoretic, must overcome the limitation recorded in Hypothesis 3.5. Arguments that overcome or sidestep parity-type limitations exist in other settings. Friedlander and Iwaniec [1] showed that $X^2 + Y^4$ represents infinitely many primes, using bilinear (Type II) estimates to overcome the parity obstruction. Maynard [4] proved that there are infinitely many primes with a restricted digit, using Fourier-analytic control of Type I and Type II sums; this is a related use of bilinear-sum methods rather than a direct parity-breaking result of the Friedlander–Iwaniec kind. Until such an ingredient is brought to bear on prime gaps, DHL[3, 2] and $H_1 \leq 6$ remain the natural limit of the GPY/Maynard–Tao method under Hypothesis 3.5.

Appendix A. Verification certificate

The formal development accompanying this note was machine-checked as follows.

- Repository:
  https://github.com/BenFrohman/DHL-3-2-implies-liminf-prime-gap-le-6
- Verified commit:
  `9a9f97dd5c9353454029789c98df411fc38fd2fe`
- Lean toolchain: `leanprover/lean4:v4.30.0-rc1`
- Mathlib: tag `v4.30.0-rc1`, commit (from `lake-manifest.json`)
  `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7`
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed):
  https://github.com/BenFrohman/DHL-3-2-implies-liminf-prime-gap-le-6/actions/runs/41718880552
- SHA-256 of the source tarball of the verified commit (`archive/9a9f97d.tar.gz` on GitHub):
  `fce92cd5198dd1cea525e468b1aa9f3f2d6726e9ffb1b6f77dc599df01a6df7e`

The tarball hash refers to the verified commit rather than to the release tag, since the tag also contains this paper and a document cannot contain the hash of an archive that includes itself. The Lean sources (`DHL32/`, `AxiomAudit.lean`, `lakefile.lean`, `lean-toolchain`, `lake-manifest.json`) are unchanged between that commit and the release tag.

The development contains no `sorry` and no user-declared axioms. Every theorem and lemma depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound`. The table below is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) from a fresh run on the verified commit.

Lean declaration	`#print axioms`
`DHL32.p_strictMono`	`[propext, Classical.choice, Quot.sound]`
`DHL32.p_prime`	`[propext, Classical.choice, Quot.sound]`
`DHL32.p_zero`	`[propext, Classical.choice, Quot.sound]`
`DHL32.p_odd`	`[propext, Classical.choice, Quot.sound]`
`DHL32.two_le_gap`	`[propext, Classical.choice, Quot.sound]`
`DHL32.gap_even`	`[propext, Classical.choice, Quot.sound]`
`DHL32.not_H1Le_one`	`[propext, Classical.choice, Quot.sound]`
`DHL32.H1Le_three_iff_two`	`[propext, Classical.choice, Quot.sound]`
`DHL32.H1Le_three_iff_twin`	`[propext, Classical.choice, Quot.sound]`
`DHL32.residues_small`	`[propext, Quot.sound]`
`DHL32.admissible_046`	`[propext, Classical.choice, Quot.sound]`
`DHL32.admissible_026`	`[propext, Classical.choice, Quot.sound]`
`DHL32.gap_le_of_primes`	`[propext, Classical.choice, Quot.sound]`
`DHL32.dhl32_H1Le_six`	`[propext, Classical.choice, Quot.sound]`
`DHL32.DHLfor_prime`	`[propext, Classical.choice, Quot.sound]`
`DHL32.Ptoy_no_twin`	`[propext, Quot.sound]`
`DHL32.Ptoy_iff_mod_four_mem`	`[propext, Classical.choice, Quot.sound]`
`DHL32.Ptoy_hit`	`[propext, Quot.sound]`
`DHL32.Ptoy_hit'`	`[propext, Quot.sound]`
`DHL32.Ptoy_DHL32`	`[propext, Classical.choice, Quot.sound]`
`DHL32.admissible_02`	`[propext, Quot.sound]`
`DHL32.dhl32_not_implies_dhl22_combinatorially`	`[propext, Classical.choice, Quot.sound]`
`DHL32.parity_two_needs_three`	`[propext, Quot.sound]`
`DHL32.admissible_three_diam_ge_six`	`[propext, Classical.choice, Quot.sound]`
`DHL32.min_diam_admissible_three`	`[propext, Classical.choice, Quot.sound]`
`DHL32.parity_floor`	`[propext, Quot.sound]`

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